CARPL DICOM Push Integration Guide for AI Algorithms

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1. Introduction

This document describes the **DICOM Push** integration model for deploying AI algorithms on the CARPL platform. In this integration model, the AI engine is exposed as a DICOM node and CARPL transmits studies directly to the AI using standard DICOM networking (C-STORE). The AI processes the received study and returns results either as structured DICOM outputs or by embedding inference metadata into DICOM objects.

Unlike file-based or API-based integrations, the DICOM Push model uses standard DICOM transport for interoperability with PACS, routers, and modality ecosystems.

Key characteristics:

- **Standards-based communication:** Studies are pushed using DICOM C-STORE.
- **Network-based integration:** CARPL acts as the DICOM SCU (sender), and the AI endpoint acts as the DICOM SCP (receiver).
- **Minimal integration overhead:** AI is integrated like any standard DICOM node.
- **Configurable routing endpoint:** Requires only AE Title, IP, and Port configuration.
- **Flexible result return options:** AI results can be returned either:
    1. As a DICOM object with probabilities in a private DICOM tag
    2. As a DICOM Structured Report (SR) in **TID 1500** format

2. DICOM Node Configuration

To enable DICOM Push integration, the AI system must expose a DICOM receiver endpoint (Storage SCP) that CARPL can route studies to.

The following configuration must be provided during setup:

Parameter	Description	Example
AE Title	DICOM Application Entity Title of the AI receiver	AI_ENGINE
IP Address	Reachable IP of the AI DICOM node	10.10.20.45
Port	Listening DICOM port of the AI receiver	104

These values must be configured in the CARPL routing page when onboarding the AI node.

Required Configuration on CARPL

CARPL must be configured with:

- Destination **AE Title**
- Destination **IP Address**
- Destination **Port**

Required Configuration on AI Endpoint

The AI DICOM listener must:

- Accept incoming C-STORE associations
- Receive complete study/series objects
- Persist incoming DICOM instances for processing
- Trigger inference once the expected study/series is fully received

3. Workflow Overview

1. Study arrives in CARPL
2. CARPL identifies routing rule for the AI algorithm
3. CARPL opens a DICOM association to the configured AI endpoint
4. CARPL pushes study instances via **C-STORE**
5. AI receives and stores DICOM objects
6. AI processes the study
7. AI returns results to the **CARPL DICOM server** using the configured CARPL DICOM destination

4. DICOM Push (Inbound to AI)

CARPL acts as a DICOM sender (SCU) and transmits the study to the AI node.

Protocol Used

- **DICOM C-STORE**
- AI acts as **Storage SCP**
- CARPL acts as **Storage SCU**

AI Requirements

The AI receiver must support:

- Association acceptance
- C-STORE reception
- Handling of study completeness
- Deduplication / retransmission safety
- Timeout handling for partial studies

The AI should not begin inference on first image arrival unless explicitly designed to do so. It is recommended to trigger processing only after full study receipt is confirmed.

5. Return of Results to CARPL

Once inference is complete, the AI must send the generated results back to the **CARPL DICOM server** over DICOM.

CARPL will expose a DICOM receiver endpoint for result ingestion, and the AI system must configure CARPL as a DICOM destination on its side.

The AI team must configure the following **CARPL DICOM server** details within their system:

Parameter	Description	Example
AE Title	CARPL DICOM AE Title for receiving results	CARPL_DICOM
IP Address	CARPL DICOM receiver IP	10.10.30.50
Port	CARPL DICOM receiver port	11112

These details will be shared by the CARPL team during integration.

AI Requirement for Result Push

The AI must:

- Configure CARPL as a DICOM destination
- Establish DICOM association to CARPL
- Push result DICOM objects back to CARPL using **C-STORE**
- Ensure results are associated with the correct input study

All AI outputs must be returned to CARPL through this configured DICOM route.

6. Result Return Methodologies

The AI result can be sent back in any of the standard DICOM output types. It can include Secondary Capture (SC), DICOM Structured Report (SR), DICOM PDF, DICOM Seg, RT-Struct. The AI can return probabilities in two supported formats which will be showcased in the deployment project table.

6.1 Method 1: Probability in Private DICOM Tag

In this approach, the AI returns a DICOM object (typically derived image, SC, or copied source instance) containing the inference output encoded in a private DICOM tag as a JSON payload.

Private Tag

- **Private Creator Tag:** (0013,0010) = "CARPL_AI"
- **Private Findings Tag:** (0013,1001) = JSON findings payload

Example

Tag	VR	Meaning	Example
(0013,0010)	LO	Private Creator	CARPL_AI
(0013,1001)	LT / UT	AI Findings JSON	JSON payload

JSON Payload Format

The findings must be encoded in the private tag as a JSON object in the following format:


```
{
  "findings": [
    { "name": "Pleural effusion", "probability": 0.87 },
    { "name": "Consolidation", "probability": 0.42 }
  ]
}
```


Notes

- (0013,0010) reserves the private block for CARPL_AI
- (0013,1001) stores the complete findings JSON payload
- The payload must be valid JSON
- probability values should be normalized to **0–100** or can also include String values
- name should contain the final finding label as agreed during integration
- Multiple findings must be included as separate objects in the findings array

6.2 Method 2: DICOM Structured Report (SR) – TID 1500

In this approach, the AI returns results as a DICOM Structured Report using **TID 1500 Measurement Report**.

TID 1500 is the preferred standards-based approach for encoding AI findings in DICOM and is recommended for interoperable deployments.

Format

- Output object type: **DICOM SR**
- Template: **TID 1500 – Measurement Report**
- Contains:
    - Observation context
    - Finding codes
    - Probability / confidence
    - Referenced images

Required Content

The SR should include:

- Observer context (AI algorithm identity)
- Referenced study/series/SOP
- Finding concept name
- Probability / confidence score

8. Networking & Connectivity Requirements

Ensure the following between CARPL and AI endpoint:

- CARPL can reach AI IP over network
- AI DICOM port is open and listening
- AI can reach CARPL DICOM server over network
- CARPL DICOM result port is open and reachable
- Firewall allows bidirectional DICOM traffic
- AE Titles are configured correctly on both ends

Recommended validation:

- DICOM C-ECHO
- DICOM C-STORE test push to AI
- DICOM C-STORE result push to CARPL
- Study completeness validation
- Result return validation